COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8/13/26	**DATE SUBMITTED:**	8/13/26
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PRELIM EXAM SKILLS	**Installing Ansible, Building an Inventory, Using Metagroups, Variables, and Creating a Playbook**

1. PROCEDURE OUTPUT

Hostname	Operating System	IP Address
villareal1@server1	Ubuntu	192.168.56.132
villareal1@server2	Ubuntu	192.168.56.135
	CentOS	

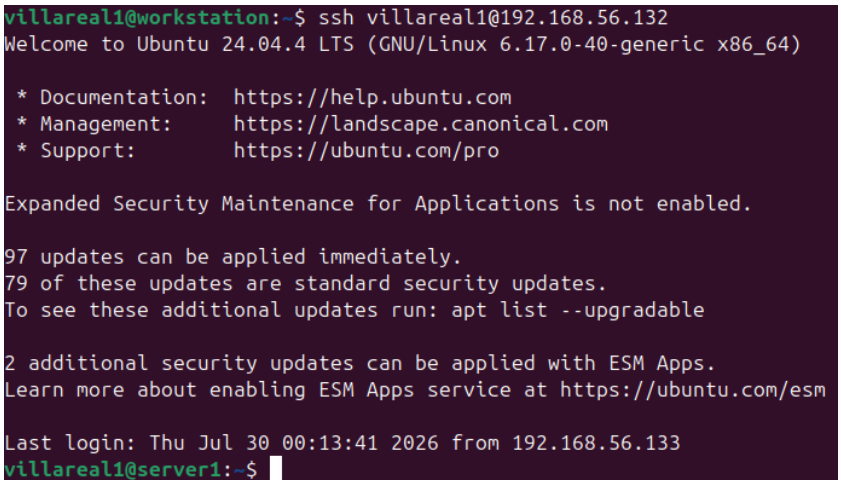
Part A: Installing and Verifying Ansible

Ansible Version

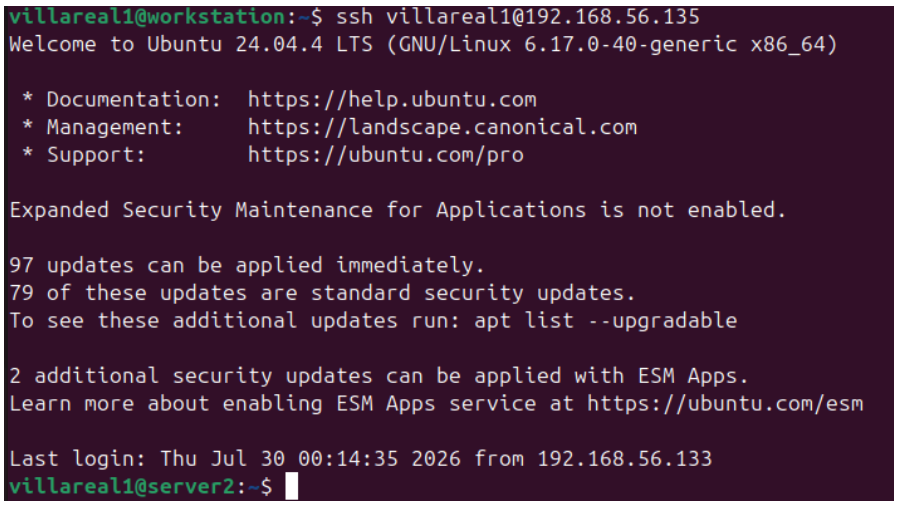
```
villareal1@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/villareal1/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/villareal1/.ansible/collections:/usr/share
/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

SSH connectivity

Server1



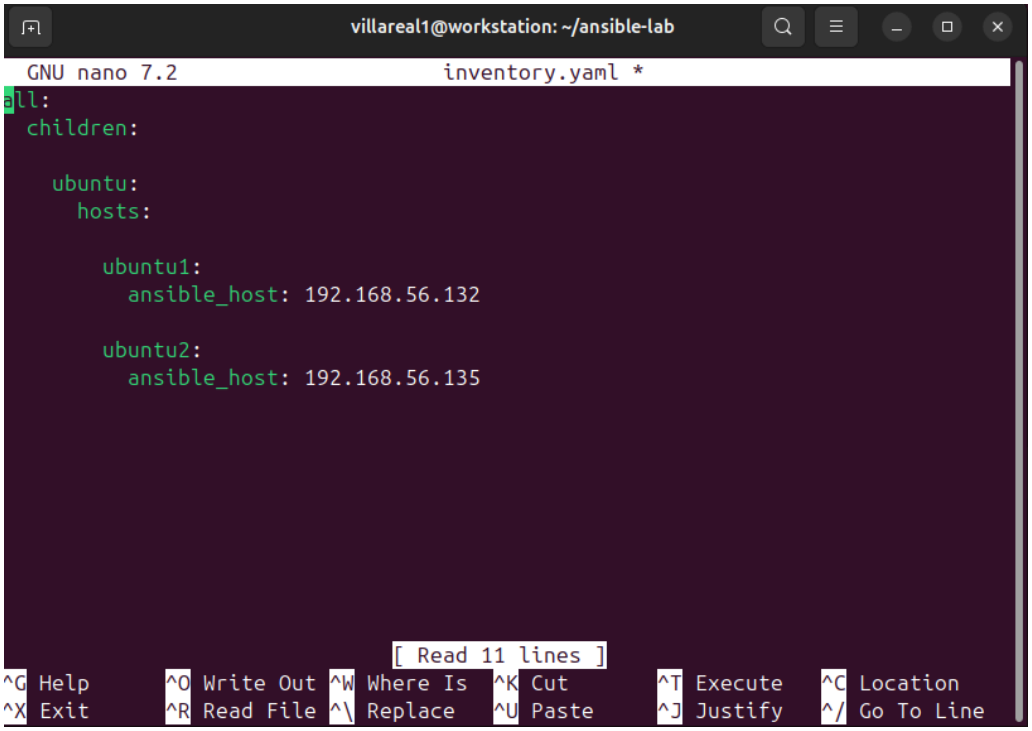
Server2



CentOS

Part B: Building an Inventory File

Inventory.yaml



Successful inventory verification

```
villareal1@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "ubuntu1": {
                "ansible_host": "192.168.56.132"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.135"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1",
            "ubuntu2"
        ]
    }
}
```

```
villareal1@workstation:~/ansible-lab$ ansible all --list-hosts
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that
the implicit localhost does not match 'all'
  hosts (0):
```

Successful ping results

```
villareal1@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup

Demonstrate successful execution of the metagroup.

```
villareal1@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.
yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
villareal1@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inven
tory.yaml
  hosts (2):
    ubuntu1
    ubuntu2
```

Part D: Creating Variables

Demonstrate successful use of variables.

```
villareal1@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvar
s.yaml

PLAY [Display Variables] ***************************************************

TASK [Gathering Facts] *****************************************************
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Show Operating System] ***********************************************
ok: [ubuntu1] => {
    "msg": "Host ubuntu1 is running Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Host ubuntu2 is running Ubuntu"
}

PLAY RECAP *****************************************************************
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Part E: Creating Your First Playbook

Demonstrate:

- **Successful playbook execution**

```
villareal1@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playboo
k.yaml

PLAY [First Ansible Playbook] *************************************************

TASK [Gathering Facts] ********************************************************
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Ping all managed nodes] *************************************************
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Display Hello World] ****************************************************
ok: [ubuntu1] => {
    "msg": "Hello World from ubuntu1"
}
ok: [ubuntu2] => {
    "msg": "Hello World from ubuntu2"
}

TASK [Display Operating System] ***********************************************
ok: [ubuntu1] => {
    "msg": "Operating System: Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Operating System: Ubuntu"
}
```

- **Successful task completion on all managed nodes**

```
PLAY RECAP ********************************************************************
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named **servers** that contains only:

- ubuntu1
- ubuntu2

villareal1@workstation: ~/ansible-lab

GNU nano 7.2 server.yaml

```
all:
  children:

    ubuntu:
      vars:
       ansible_user: villareal1
      hosts:

        ubuntu1:
          ansible_host: 192.168.56.132
          os_type: Ubuntu

        ubuntu2:
          ansible_host: 192.168.56.135
          os_type: Ubuntu

    linuxservers:
      children:
        ubuntu:
```

Create a playbook named challenge.yaml that displays the message:

None

```
Welcome to Ansible Laboratory
```

villareal1@workstation: ~/ansible-lab

GNU nano 7.2 challenge.yaml

```
---
- name: First Ansible Playbook
  hosts: linuxservers
  tasks:
    - name: Display Greeting
      debug:
        msg: "Welcome to Ansible Laboratory"
```

Run the playbook and demonstrate that only hosts in the servers group receive the message

```
villareal1@workstation:~/ansible-lab$ ansible-playbook -i server.yaml challenge.
yaml

PLAY [First Ansible Playbook] *************************************************

TASK [Gathering Facts] ********************************************************
ok: [ubuntu1]
ok: [ubuntu2]

TASK [Display Greeting] *******************************************************
ok: [ubuntu1] => {
    "msg": "Welcome to Ansible Laboratory"
}
ok: [ubuntu2] => {
    "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP ********************************************************************
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Guide Questions

1. What is the role of the Control Node in Ansible?
   - The control node is where you install the ansible and is used for automating tasks, playbooks and executing commands.
2. Why is an inventory file required?
   - The inventory file's purpose is to list all of the managed nodes in order for ansible to manage all servers at the same time in one command.
3. What is the purpose of a metagroup?
   - The metagroup is basically the organizer of multiple smaller groups.
4. How do variables simplify Ansible administration?
   - Variables in ansible are used for simplifying since once created it can be reused.
5. What is the difference between an ad hoc command and a playbook?
   - Ad hoc command is only used for single line command while playbook is for scripting similar to bash.

NOTE: CentOS IS NOT INSTALLED ON THE USED DEVICE THEREFORE NO MANAGED NODE CENTOS IS USED FOR THIS SKILLS EXAM.

2.1. CONCLUSIONS/LEARNINGS:

In conclusion, ansible's default amount of control nodes and managed nodes has 1 each and this is enough to configure as long as you can ssh command on the control node into the managed node. Ansible is useful for automating simple tasks and displaying messages for the user.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Idea generation and explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to generate ideas and explain unexplored concepts from ansible.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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